SOLUTION
SECTION-I

Single Correct Answer Type **19 Q. [3 M (–1)]**

1. Ans. (C)

Sol. Least count = 1MSD – 1VSD = 0.5 mm $-\left(\frac{49}{50}\right)$ (0.5 mm) = 0.001 cm

Reading = 7.45 + 29 × 0.001 = 7.479 cm

2. Ans. (C)

Sol. $m = \frac{F}{a} = 1kg$

$$\frac{dm}{m} = \frac{dF}{F} + \frac{da}{a}$$

$$= \frac{0.1}{2} + \frac{0.1}{2} = 0.1$$

3. Ans. (B)

Sol. Least count = $\frac{\text{pitch}}{\text{No. of divisions}} = \frac{2}{200} = 0.01\text{mm}$

Reading = 3× 2 + (46–5) (0.01) = 6.41 mm

4. Ans. (B)

Sol. Least count = $\frac{\text{pitch}}{\text{No. of divisions}} = \frac{0.5}{50} = 0.01$ mm

Diameter of wire = 6 × 0.5 + 46 × 0.01 = 3.46 mm

5. Ans. (A)

Sol. From screw gauge z.e. = + 0.20 mm
Dia of chalk = 3.5 – 0.2 = 3.3 mm = 0.33 cm
For vernier callipers, z.e. = – 0.05 cm
Thus it must show 0.33 – 0.05 = 0.28 cm
so (a) is correct

6. Ans. (B)

7. Ans. (A)

Sol. $F_B = W - W'$
$= 1 \pm (0.1)$

$$\frac{\rho_0 Vg}{\rho_w Vg} = \frac{5 \pm 0.05}{1 \pm 0.1}$$

$$\frac{\rho_0}{\rho_w} = \frac{5 \pm 0.05}{1 \pm 0.1}$$

$= (5) \pm (\Delta\rho)$

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$$\frac{\Delta\rho}{5}=\frac{0.05}{5}+\frac{0.1}{1}$$

$= 0.01 + 0.1 = 0.11$

8. Ans. (C)

Sol. $$\frac{\Delta u}{u}\times 100=\left(\frac{2\Delta d}{d}+\frac{\Delta v}{v}+\frac{\Delta \ell}{\ell}\right)\times 100$$

$$\frac{\Delta \ell}{\ell}=\frac{30}{1800}$$

$$=\left[2\times\frac{0.04}{1-6}+\left(\frac{30}{1800}\right)\right]\times 100$$

$= 8.166\% \approx 8.2\%$

9. Ans. (C)

Sol. $$\text{Least count} = \frac{\text{Pitch}}{\text{Number of division of circular scale}}$$

$$=\frac{1\text{mm}}{100}=0.01\text{mm}=0.001\text{cm}$$

Zero error = –0.03 mm

Measurement = 3.76 – (–0.03) = 3.79 mm

10. Ans. (C)

Sol. Chritina and Alice both measure error of 2 mm while bob measures error of 4 mm.

11. Ans. (C)

Sol. $$\frac{1}{2f\sqrt{\mu}}=8\times10^{-3}$$

$$\frac{1}{2\times10^{-2}\times8\times10^{-3}}=f$$

$$f=\frac{10^5}{16}=\frac{100\times100\times10}{4\times4}$$

$= 6250$ Hz

$$\frac{\Delta f}{f}=\frac{1}{2}\frac{\Delta\mu}{\mu}+\frac{\Delta\text{slope}}{\text{slope}}$$

$$=\frac{1}{10}+\frac{0.3}{80}=\frac{11}{80}$$

$$\Delta f=6250\times\frac{11}{80}=859.8\text{Hz}$$

12. Ans. (D)

Sol. $R = \sqrt{AB}$

$R = r \pm \Delta r$

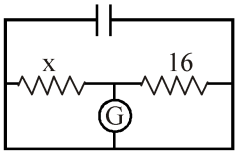
where, $r = \sqrt{ab}$

and $$\frac{\Delta r}{r}=\frac{1}{2}\frac{\Delta a}{a}+\frac{1}{2}\frac{\Delta b}{b}$$

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13. Ans. (B)

Sol.



$$\frac{x}{16}=\frac{36}{64}=9\Omega$$

$$\frac{dx}{x}=\frac{d\ell}{\ell(1-\ell)}=\frac{0.5\times10^{-1}}{36\times64}$$

$$dx=\frac{0.05\times9}{36\times64}=\frac{1}{5120}\Omega$$

$$x=9\pm\frac{1}{5120}\Omega$$

14. Ans. (B)

Sol. 1MSD = 1 mm, 1VSD = 0.4 mm L.C = 0.2 mm

Zero error = $-(5\times0.4-1)$mm = -1mm

Reading = $22+(1-2\times0.4)$ – zero error

$= 22+0.2+1$

$= 23.2$ mm

$\Rightarrow$ report = $(23.2\pm0.2$ mm$)$

15. Ans. (B)

Sol. $$Y=\frac{F/A}{\Delta L/L}$$

$$Y=\frac{10/\pi\left(4\times10^{-8}\right)}{0.8\times10^{-3}/2}$$

$$=\frac{20}{0.8\times4\times\pi\times10^{-11}\times2}$$

$$=2\times10^{11}$$

$$\frac{\Delta y}{y}=2\frac{\Delta d}{d}+\frac{\text{Error}(\Delta L)}{\Delta L}$$

$$=2\frac{0.01}{0.4}+\frac{0.04}{0.8}$$

$$==\frac{0.08}{0.8}=0.1$$

16. Ans. (D)

Sol. Lesser the least count, more precise the instrument is

$$LC_1=\frac{1}{200},\ LC_2=\frac{0.5}{400},\ LC_3=0.1\text{ mm},\ LC_4=0.2\text{ mm}$$

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17. Ans. (C)

Sol. Voltmeter is connected in parallel & Ammeter in series as shown in option (C).

18. Ans. (B)

Sol. $r = \left(\frac{1-a}{1+a}\right)$

$$\frac{\Delta r}{r} = \frac{\Delta(1-a)}{(1-a)} + \frac{\Delta(1+a)}{(1+a)}$$

$$= \frac{\Delta a}{(1-a)} + \frac{\Delta a}{(1+a)}$$

$$= \frac{\Delta a(1+a+1-a)}{(1-a)(1+a)}$$

$$\therefore \Delta r = \frac{2\Delta a}{(1-a)(1+a)}\frac{(1-a)}{(1+a)} = \frac{2\Delta a}{(1+a)^2}$$

19. Ans. (C)

Sol. $N = N_0 e^{-\lambda t}$

$\ell nN = \ell nN_0 - \lambda t$

$$\frac{dN}{N} = -d\lambda t$$

Converting to error,

$$\frac{\Delta N}{N} = \Delta\lambda t$$

$$\therefore \Delta\lambda = \frac{40}{2000 \times L} = 0.02$$ (N is number of nuclei left undecayed)

Multiple Correct Answer Type — 5 Q. [4 M (–1)]

20. Ans. (B,D)

Sol. $\ell_1 + e = \frac{\lambda}{4}$ Case-1 : $\frac{\Delta v}{v}\% = \frac{0.1 \times 100}{33} \approx 0.3\%$

$\ell_2 + e = \frac{3\lambda}{4}$ $e = \frac{50-51}{2} = -ve$

$\lambda = 2(\ell_2 - \ell_1)$ Case-2 : $\frac{\Delta v}{v}\% = \frac{0.1 \times 100}{50.8} \approx 0.2\%$

$\Rightarrow v = f\lambda$ $\Rightarrow$ $e = \frac{76-75.6}{2} = 0.2 = +ve$

$\frac{\Delta v}{v} = \frac{(\Delta\ell_2 + \Delta\ell_1)}{\ell_2 - \ell_1}$ Case-3 : $\frac{\Delta v}{v}\% = \frac{0.1 \times 100}{30.4} \approx 0.33\%$

$e = \frac{\ell_2 - 3\ell_1}{2}$ $e = \frac{46.0 - 46.8}{2} = -ve$

Case-4 : $\frac{\Delta v}{v}\% = \frac{0.1 \times 100}{31.5} \approx 0.32\%$

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$$e = \frac{48.2 - 48.0}{2} = 0.1\text{cm}$$

21. Ans. (A, C)

Sol. Using lens formula and data given in graph.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \qquad ...(1)$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{-10}$$

$f = +5$

By differentiating eq. (1)

$$\Rightarrow \frac{-\Delta f}{f^2} = \frac{-\Delta v}{V^2} + \frac{\Delta u}{u^2} = \frac{1}{+V^2}\Delta v + \frac{1}{u^2}\Delta u +$$

$$\Rightarrow + \frac{\Delta f}{5^2} = \frac{1\times(0.1)}{+10^2} + \frac{1\times(0.1)}{10^2}$$

$$\Delta f = \frac{0.2}{100}\times 25 = \frac{0.2}{4} = 0.05$$

so, $f = 5 \pm 0.05$

Fractional error is 0.01

22. Ans. (B, C)

Sol. Percentage errors in calculation of a, b and M are

$$\frac{\Delta a}{a}\times 100\% = \frac{0.01}{4}\times 100\% = 0.25\%$$

$$\frac{\Delta b}{b}\times 100\% = \frac{0.01}{2}\times 100\% = 0.5\%$$

$$\frac{\Delta M}{M}\times 100\% = \frac{0.003}{0.3}\times 100\% = 1\%$$

Moment of inertia about the given axis

$$I = \frac{1}{2}M\left(a^2 + b^2\right)$$

$$\frac{\Delta I}{I} = \frac{\Delta M}{M} + \frac{2a\Delta a + 2b\Delta b}{a^2 + b^2}$$

$$= 0.01 + \frac{2(4)(0.01) + 2(2)(0.01)}{20}$$

$$\frac{\Delta I}{I}\times 100\% = 1 + 5\{0.08 + 0.04\} = 1.6\%$$

23. Ans. (A, C)

Sol. Let be λ is wavelength of sound in air

$$\ell_1 = \frac{\lambda}{4} \text{ and } \ell_2 = \frac{3\lambda}{4}$$

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$\lambda = 2(\ell_2 - \ell_1)$

$v = f\lambda = 165 \times 2(150 - 50) \times 10^{-2} = 330$ m/s

$$\frac{dv}{v} = \frac{|d\ell_2| + |d\ell_1|}{\ell_2 - \ell_1}$$

$$dv = \frac{0.5 + 0.1}{150 - 50} \times 330 = 1.98 \text{ m/s}$$

$v = (330 \pm 2)$ m/s, it is speed of sound in air.

24. Ans. (A,C,D)

Sol. $$\frac{R_x}{x} = \frac{R}{1-x}$$

$$\frac{R_x}{20} = \frac{20}{80}$$

$R_x = 5$

$$R_x = \frac{x}{1-x}[R]$$

$$\Delta R_x = R\left[\frac{\Delta x(1-x) - x(-\Delta x)}{(1-x)^2}\right]$$

$\Delta R_x = 0.03\Omega$

error will be minimused when null-point is found at middle of wire.

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 5 Q. [4 M (0)]

25. Ans. 0

Sol. $$dv = -\frac{1}{2}\frac{v_0 T_0^{1/2} e^{\left[\frac{2(T-T_0)}{3\ T_0}\right]}}{T^{3/2}}dT + v_0\left(\frac{T_0}{T}\right)^{1/2} e^{\left[\frac{2(T-T_0)}{3\ T_0}\right]} = \frac{2}{3T_0}dT$$

$$\frac{dv}{v} = -\frac{1}{2}\frac{dT}{T} + \frac{2dT}{3T_0} = \left(\frac{2}{3T_0} - \frac{1}{2T}\right)dT$$

At $T = \frac{3T_0}{4}$, $\frac{dv}{v} = 0$

$\therefore$ % error = 0

26. Ans. 7

Sol. $(7 \pm 1) \times 10^{-3}$ T

27. Ans. 4

Sol. $f = \frac{v}{2(\ell + 2e)}$ where e = end correction = 0.6 r

$$\therefore f = \frac{v}{2(\ell + 2\times 0.6r)} = \frac{v}{2(\ell + 1.2r)} \therefore \frac{\Delta f}{f} = \frac{\Delta v}{v} - \frac{\Delta(\ell + 1.2r)}{\ell + 1.2r} = \frac{\Delta v}{v} - \frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r}$$

here $\frac{\Delta v}{v} = 0$ (given) $\frac{\Delta f}{f} \times 100 = -\frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r} \times 100$

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for maximum % error : $\Delta\ell = 0.1$, $\Delta r = 0.05$

$$\left(\frac{\Delta f}{f}\times 100\right)_{max} = \frac{0.1+1.2\times 0.05}{94+1.2\times 5}\times 100 = \mathbf{0.16\%}$$

28. Ans. 1

Sol. $\tan\theta = \dfrac{r\omega^2}{g} = \dfrac{R(\sin\theta)\omega^2}{g}$

or $\cos\theta = \dfrac{g}{R\omega^2}$, i.e., $\dfrac{R-h}{h} = \dfrac{g}{R\omega^2}$

$h = R - (g/\omega^2)$

$g = \omega^2(R - h)$

$dg = -\omega^2 dh$, i.e., $(\Delta g)_{min} = -(\omega^2)_{min}(\Delta h)_{min}$

So substituting the value of ω_{min} and $(\Delta h)_{min}$

$(\Delta g)_{min} = -100 \times 10^{-4}$ m/s^2

29. Ans. 3

Sol. $D = (1.25 \text{ mm} \pm 0.01 \text{ mm})$

$$Y = \frac{FL}{A\ell}$$

ℓ = elongation.

$$\%\text{ error in } Y = \frac{\Delta L}{L}\times 100 + \frac{2\Delta D}{D}\times 100 + \frac{\Delta\ell}{\ell}\times 100$$

$$= \left(\frac{0.01}{100}\times 100\right) + \left(\frac{0.02}{1.25}\times 100\right) + \left(\frac{0.001}{0.125}\times 100\right) = 2.41$$

n = 3.

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